



POE- PART 1

PLANNING & DESIGN

OPEN – SOURCE CODING
OPSC6312



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YOUTHLINK SA

“YOUR NEXT OPPORTUNITY STARTS HERE.”

Introduction

YouthLink SA is a proposed Android application designed to help South African youth find opportunities for education, skills development and employment. The idea was developed from the research conducted on SAYouth, LinkedIn and Youth Jobs South Africa, where useful features and areas for improvement were identified.

The aim of YouthLink SA is to bring different opportunity areas together in one easy-to-use application. The application will also include features developed from gaps identified during the research, such as career exploration, support for users without matric and personalised opportunity matching.

This document explains the planned design and development of YouthLink SA, including its requirements, user interface, navigation, API, database, UML diagrams and project plan.

Overview

App Name: *YouthLink SA*

Slogan: “Your next opportunity starts here.”

App Icon Design:



Figure 1: YouthLink SA app icon. Source: Generated with assistance from ChatGPT (OpenAI, 2026).

Colours: Green (growth and opportunity), Yellow/Gold (hope and new possibilities), Red (energy and ambition), Blue (trust and confidence), White (clarity and accessibility). The interconnected pathway symbol represents **connecting young people to opportunities**, while the road leading forward represents **career and personal development**. The diverse figures represent **youth, community and inclusion**. The slogan, “Your next opportunity starts here,” reinforces the app’s purpose of helping young people discover and access opportunities.

Target Users:

YouthLink SA will mainly be aimed at South African youth, including:

-  High-school learners exploring career options
-  School leavers
-  University and TVET students
-  Graduates looking for entry-level opportunities
-  Unemployed young people
-  Young people looking for internships, WIL or learnerships
-  Users who do not have matric
-  Young people interested in entrepreneurship or volunteering.

Purpose of the Application

The purpose of YouthLink SA is to make it easier for young people to find opportunities that match their situation and interests. Instead of having to search through different websites and platforms, users will be able to find different types of opportunities in one application.

These opportunities will include:

-  Jobs
-  Entry-level jobs
-  Internships
-  WIL
-  Learnerships
-  Graduate programmes
-  Bursaries
-  NSFAS
-  University and TVET opportunities
-  Skills programmes
-  Short courses
-  Volunteering
-  Entrepreneurship
-  Youth programmes
-  Career events

How the Application Will Work:

When users register, they will provide information such as their education level, skills, interests and location. This information will help YouthLink SA suggest relevant opportunities.

However, users will not be restricted to their original choices. They can search for other fields at any time. For example, a user who initially selects IT can later search for Business and may discover opportunities related to Information Systems or Business Information Systems.

This makes the application flexible and allows users to explore different career options.

Innovative Features

YouthLink SA will combine ideas from existing platforms while adding features based on gaps identified during the research. The main difference is that the application is designed around the whole youth opportunity journey, rather than only job searching.

Important features include:

- ✚ Career exploration for learners who are unsure what they want to study.
- ✚ Opportunity matching based on skills, interests and education.
- ✚ Support for users who do not have matric.
- ✚ Study and funding information alongside employment opportunities.
- ✚ A beginner-friendly CV builder.
- ✚ English, Afrikaans and isiXhosa language options.
- ✚ Search and filters that allow users to change their interests at any time.

Detailed Application Requirements

The following requirements describe what YouthLink SA must be able to do.

Requirement	Description
Registration and Login	Users must be able to create an account and securely log in.
User Profile	Users must be able to add and edit their education, skills, interests, experience, location and opportunity preferences.
Opportunity Search	Users must be able to search using keywords such as IT, business, finance, healthcare or marketing.

Requirement	Description
Opportunity Filters	Users must be able to filter by location, field, opportunity type, education level, experience, closing date and remote/in-person options.
Career Explorer	Users must be able to explore careers and see simple information about qualifications, possible jobs and study options.
Study and Funding	Users must be able to find university, TVET, bursary, NSFAS, skills programme and short-course information.
Employment Opportunities	The app must display jobs, entry-level positions, internships, WIL, learnerships and graduate programmes.
CV/Profile Builder	Users must be able to create a basic CV using personal information, education, skills, experience and certificates.
Opportunity Matching	The system should recommend opportunities based on information in the user's profile.
No-Matric Opportunities	The app should identify suitable opportunities where matric is not required and consider skills and experience.
Saved Opportunities	Users must be able to save opportunities and access them later.
Notifications	Users should receive alerts about suitable opportunities and upcoming closing dates.
Events and Programmes	Users should be able to view career fairs, open days, workshops, youth programmes and entrepreneurship events.
Languages	The application must support English, Afrikaans and isiXhosa.
Search Flexibility	Users must be able to change their interests and search for fields that were not selected during registration.
API Connection	The Android application must communicate with a REST API to send and receive application data.
Database	User, opportunity, career, event and saved-opportunity data must be stored in a database.

Requirement

Description

Offline Storage

Important information can be temporarily stored on the device so the app can still provide selected information when there is limited connectivity.

User Interface Design

YouthLink SA will use a simple and clean interface. The aim is to make the application easy for a young person to understand without displaying too much information at once.

Main Screens

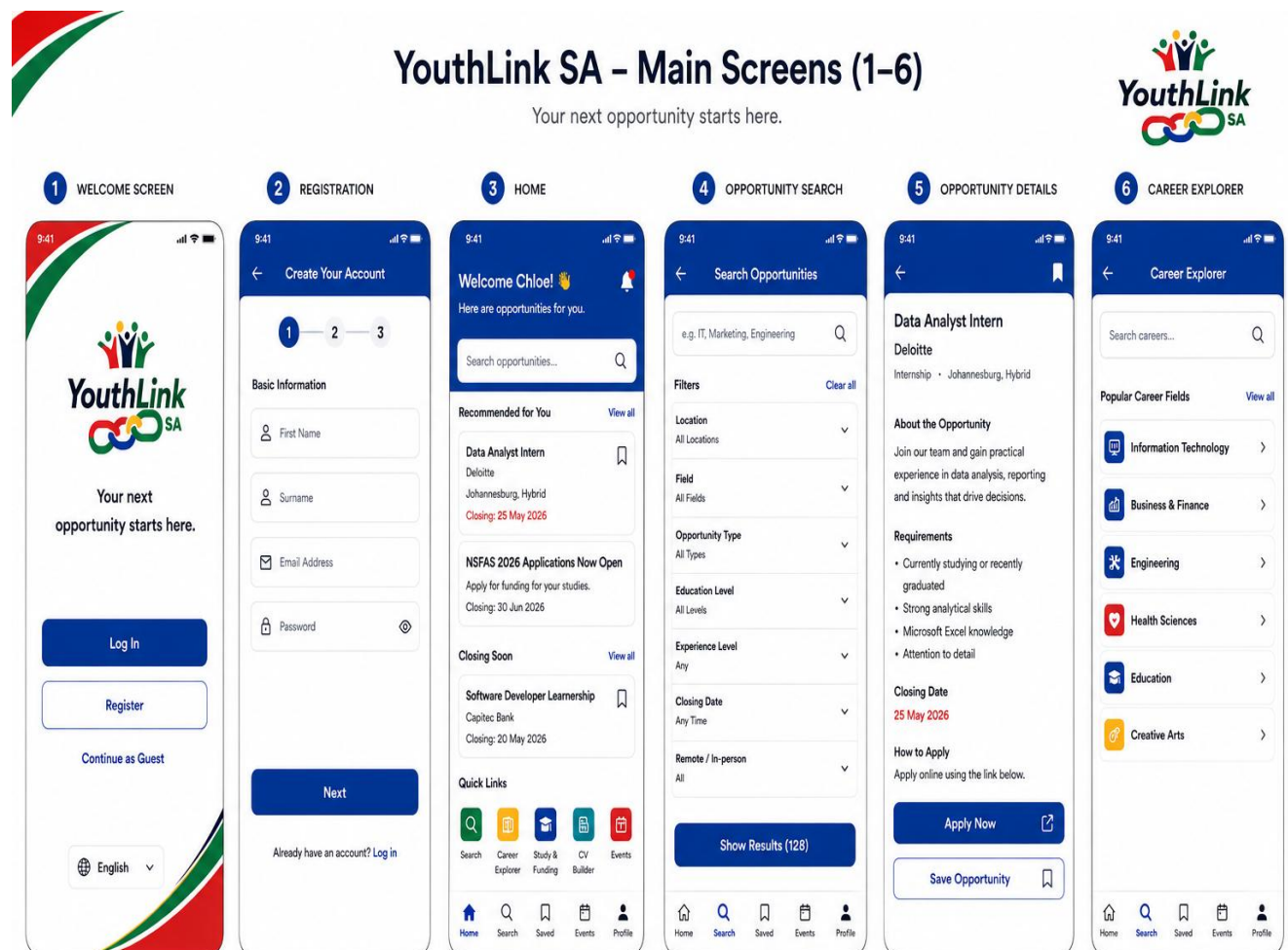


Figure 2: Main Screens (1-6) Source: Generated with assistance from ChatGPT (OpenAI, 2026).

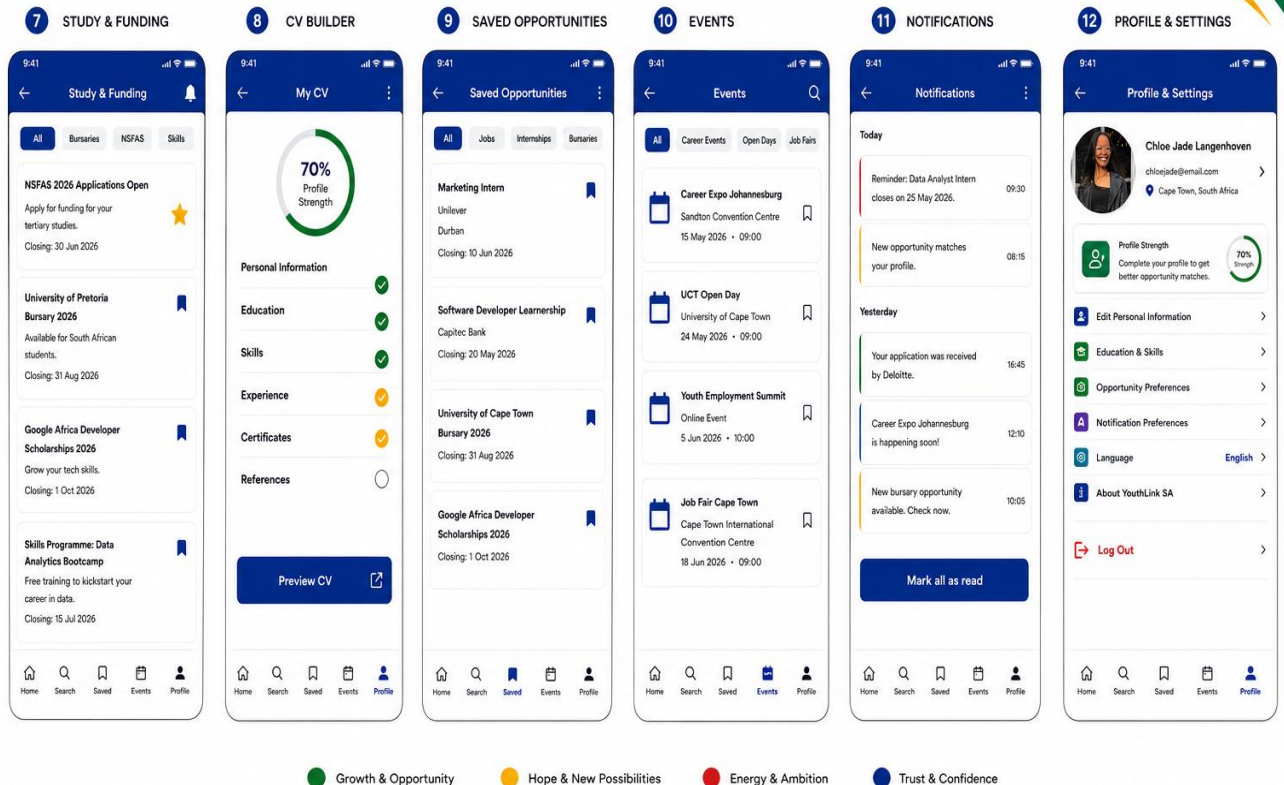


Figure 3: Main Screens (7-12) Source: Generated with assistance from ChatGPT (OpenAI, 2026).

Screen	Purpose
Welcome Screen	Introduces YouthLink SA and allows users to log in or register.
Registration	Collects the user's basic information, education, interests and skills.
Home	Shows recommended opportunities, search, closing dates and shortcuts to main features.
Opportunity Search	Allows users to search and filter opportunities.
Opportunity Details	Displays the opportunity description, requirements, closing date and application information.
Career Explorer	Allows users to explore different career fields and possible study paths.

Screen	Purpose
Study & Funding	Displays study opportunities, bursaries, NSFAS and skills programmes.
CV Builder	Allows users to create and update their CV/profile.
Saved Opportunities	Displays opportunities saved by the user.
Events	Shows career events, open days, job fairs and youth programmes.
Notifications	Displays alerts and reminders.
Profile & Settings	Allows users to update their information, language and notification preferences.

Navigation Flow

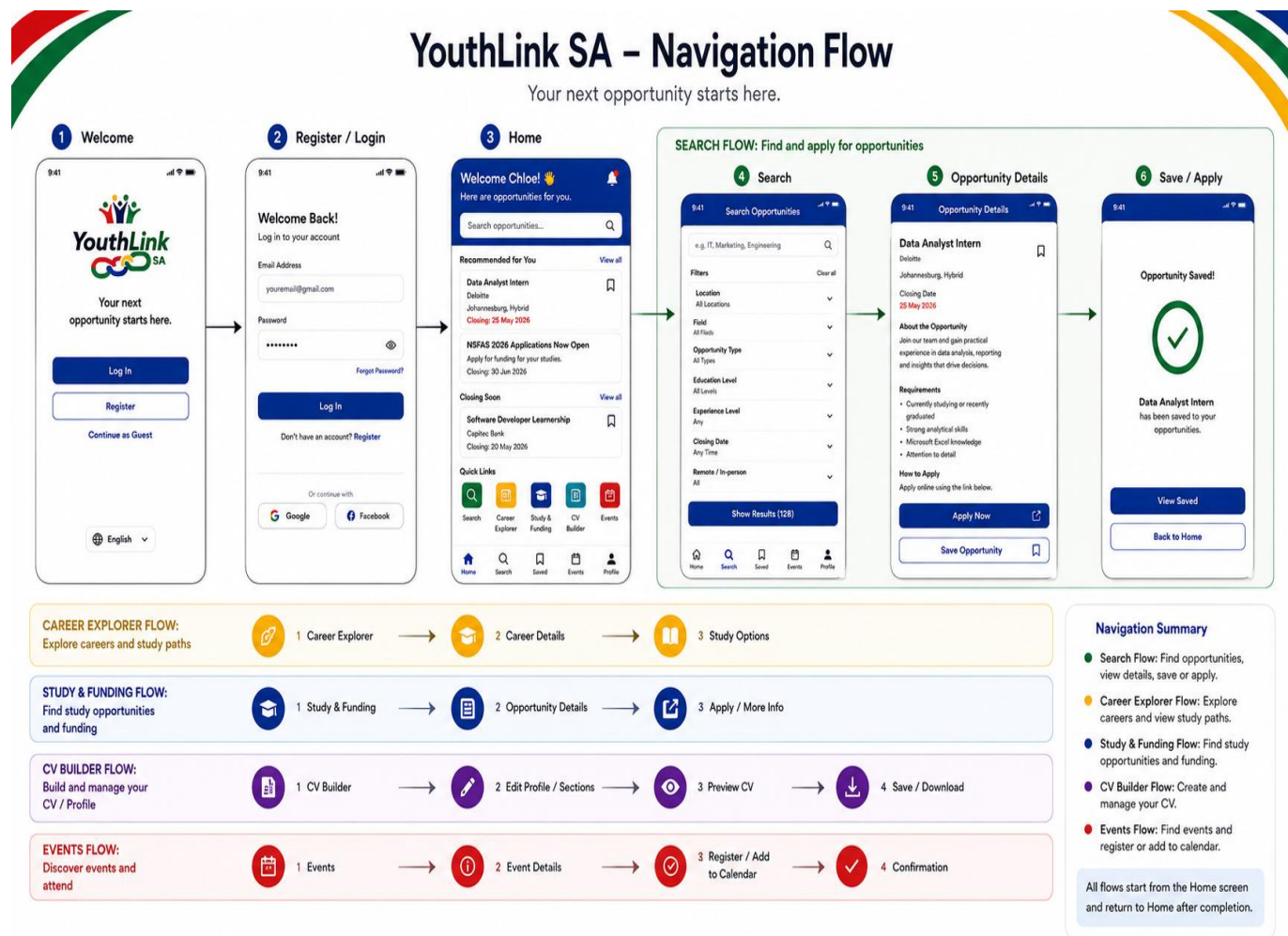


Figure 4: Navigation Flow

Source: Generated with assistance from ChatGPT (OpenAI, 2026).

API Design

YouthLink SA will use a REST API to allow the Android application to communicate with the backend.

The API will be responsible for:

- 🚦 Registering and managing users
- 🚦 Retrieving opportunities
- 🚦 Searching and filtering opportunities
- 🚦 Saving opportunities
- 🚦 Updating profiles
- 🚦 Retrieving career information
- 🚦 Retrieving study and funding information
- 🚦 Retrieving events

Managing user preferences

The Android application will send requests to the API using HTTPS. The API will process the request and communicate with the database. Information will be returned to the application using JSON.

The proposed API will be developed using Node.js and Express. The API can be hosted using a cloud service such as Render.

A MySQL database will be used to store the main application data.

For notifications, Firebase Cloud Messaging (FCM) can be used to send alerts to users.

For local/offline storage, Room Database can be used within the Android application to temporarily store selected information.

UML and Data Design

YOUTHLINK SA – UML DIAGRAMS

The UML diagrams show how the different parts of YouthLink SA communicate.

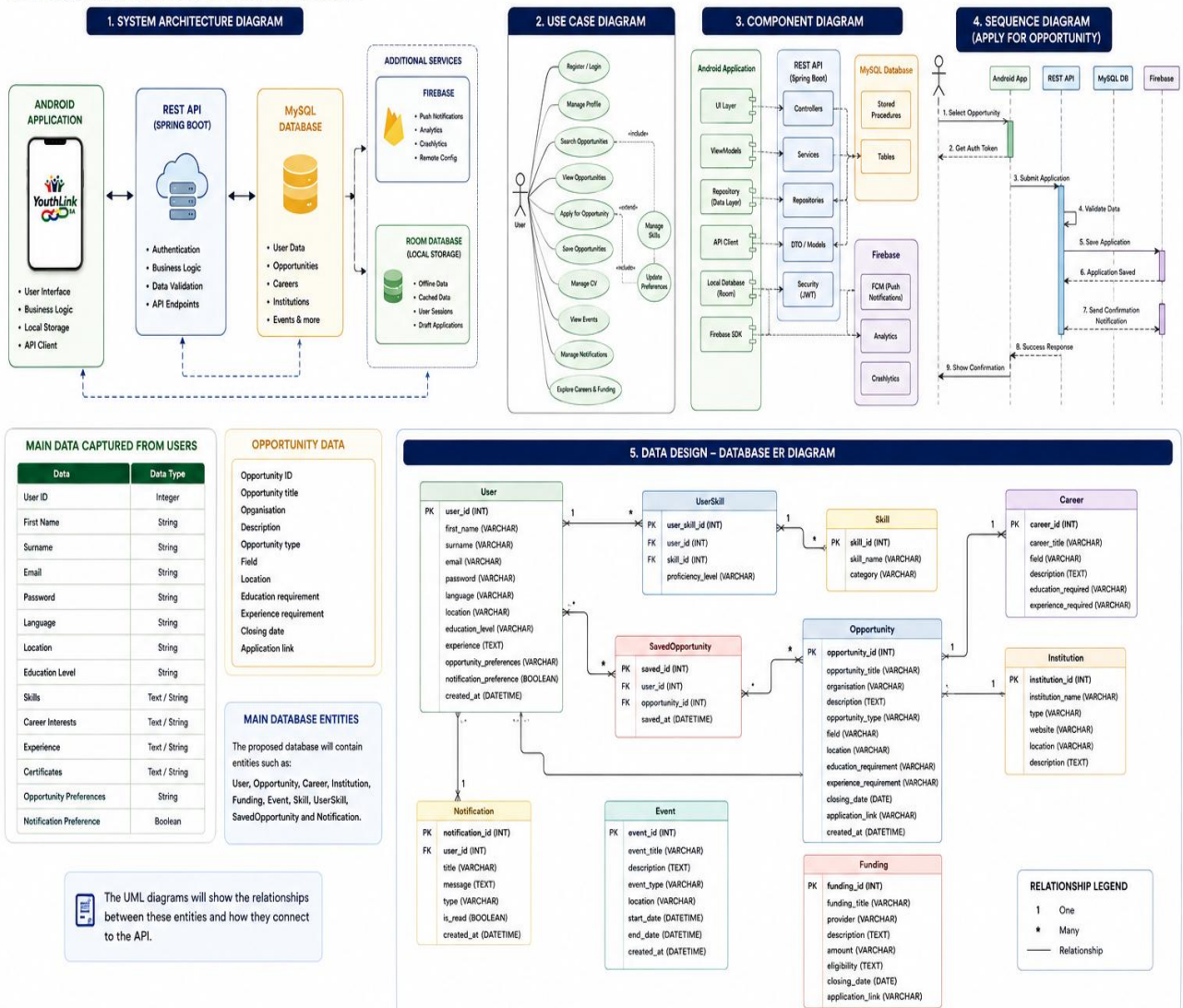


Figure 5: UML Diagrams and the Data Design

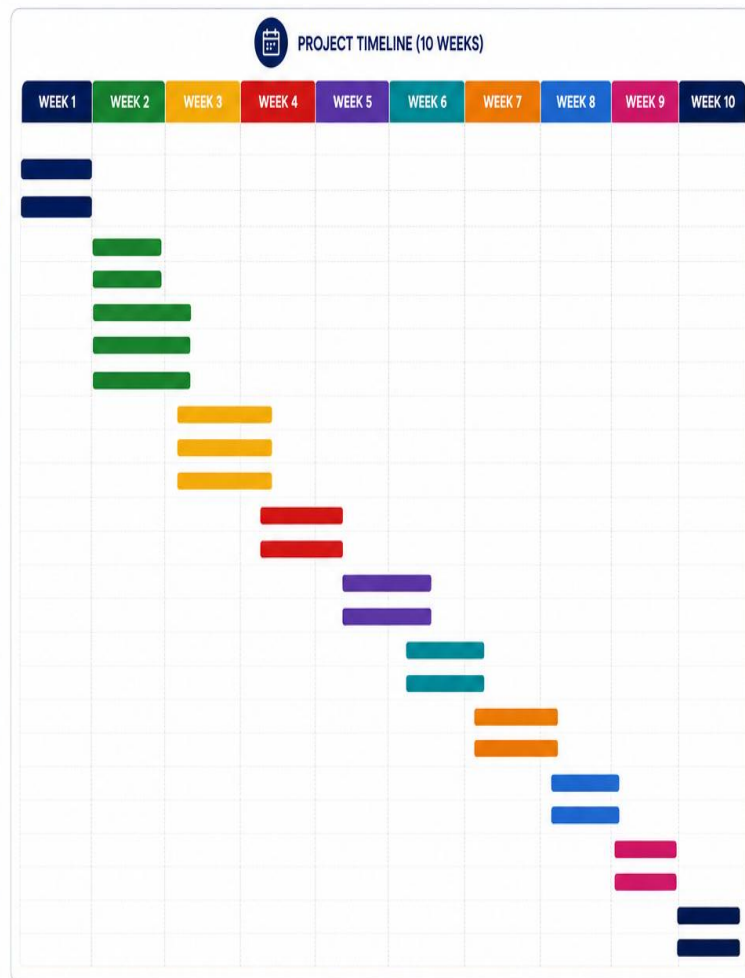
The UML and data design illustrates how the main components of YouthLink SA communicate and how application data is organised. The Android application communicates with the REST API through HTTPS, while the API manages the application logic and connects to the MySQL database. Firebase is used for services such as notifications, while Room Database provides local storage on the device. The database contains key entities such as users, opportunities, careers, institutions, funding, events, skills, saved opportunities and notifications. The relationships between these entities allow YouthLink SA to store user information, match users with opportunities, save opportunities and manage career, study and event information.

Project Plan

YOUTHLINK SA - PROJECT PLAN

The project will be divided into smaller tasks to make development easier to manage and ensure that the YouthLink SA app is delivered on time and meets all user needs.

#		TASK	PLANNED TIME
1		Research existing platforms	Week 1
2		Finalise requirements	Week 1
3		Design app branding and icon	Week 2
4		Create screen mockups	Week 2
5		Create navigation diagram	Week 2
6		Design database	Week 3
7		Design API	Week 3
8		Create UML diagrams	Week 3
9		Set up Android Studio project	Week 4
10		Create registration and login	Week 4
11		Develop main screens	Week 5
12		Develop search and filtering	Week 5
13		Develop CV/Profile Builder	Week 6
14		Develop Career Explorer	Week 6
15		Develop Study & Funding section	Week 7
16		Connect API and database	Week 7
17		Add notifications and languages	Week 8
18		Complete remaining features	Week 8
19		Testing	Week 9
20		Bug fixing	Week 9
21		Final improvements	Week 10
22		Final documentation and submission	Week 10



PROJECT GOAL

To develop a robust, user-friendly and impactful mobile application that connects South African youth with opportunities in careers, studies, funding and personal development.

- Planning & Research
- Design & Planning
- System Design
- Development Setup
- Core Development
- Feature Development
- Integration
- Enhancements
- Testing & Quality
- Finalisation

Figure 6: Project Plan

Source: Generated with assistance from ChatGPT (OpenAI, 2026).

Conclusion

The Planning and Design stage turns the findings from the research into a clear plan for developing YouthLink SA. The research on SAYouth, LinkedIn and Youth Jobs South Africa helped identify useful features and areas where YouthLink SA can provide additional value.

The proposed design brings career exploration, study and funding, employment opportunities, CV building and opportunity matching together in one application. It also considers users who may have limited experience or no matric.

The requirements, interface designs, API, database structure, UML diagrams and project plan provide a clear guide for development. These designs will be used in **Part 2** to build and test the YouthLink SA application.

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